

Fictional case study for training. Not a real investigation.



Investigation report

Rejected take-off and runway overrun, Tarlton Springs - ATSB-style
final report

Drafted 05/10/2026

INVESTIGATION WORKFLOW SUITE

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Investigation summary

What happened

On 18/06/2026, Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, was departing runway 09 at Tarlton Springs Aerodrome on the day's last scheduled sector with 44 passengers, 2 flight crew and 1 cabin crew. The roll began at 17:30 local, 5 minutes after the end of civil twilight. At 07:30:31 UTC, after V1 had been called at 112 kt, the first officer saw two kangaroos moving onto the runway from the north. The pilot in command rejected the take-off at 118 kt. One second later the No. 2 propeller and right main gear struck both animals, and No. 2 reverse was lost.

Maximum braking was established 2 seconds after the decision, faster than the operator's fleet mean of 3.2 seconds. The aircraft left the sealed surface at 21 kt and stopped upright on the extended centreline, 62 m into the 90 m runway end safety area. There were no serious injuries, although one cabin crew member reported bruising during disembarkation. The aircraft was substantially damaged. Three No. 2 propeller blades were beyond repairable limits, the right main gear doors were distorted, both right main tyres were deflated and the lower right fuselage was punctured. Both kangaroos were killed. The aerodrome was closed for 4 hours 20 minutes.

What the investigation found

The reject began 6 kt and 3 seconds after V1, consuming the 67 m margin in the scheduled accelerate-stop distance. With reverse on No. 1 only, deceleration was approximately 15% below the certified value.

The animals very likely entered through a failed section of the northern perimeter fence, whose repair had been deferred, unfunded, for three budget cycles. An animal pad through it lay 138 m before the first carcass. The aerodrome's sighting log showed most kangaroo movement in the hour after last light, when the departure took place, and nothing on the unlit strip could be seen.

No runway inspection was requested or prompted by procedure. The operator required inspections only at aerodromes in Appendix 8A, and listing required a reported wildlife strike. Tarlton Springs was never listed, despite eleven logged kangaroo sightings on the sealed surface. The evening service was introduced as a schedule change without reopening a route risk assessment restricting the aerodrome to daylight operations. A review might have addressed only the lighting and fuel matters that restriction named.

The aerodrome fell below the Aerodrome Standard AS-139 size threshold, so it needed no documented wildlife hazard management plan. This likely contributed by leaving its wildlife measures without an owner or audit trail, though that influence was less firmly established than the other factors. Rejected take-off training covered only aircraft system failures, leaving the pilot in command no prepared frame for a reject above V1. He also stated that he knew the procedure and departed from it deliberately.

Two cabin floor proximity escape path segments were dark from a failed common ballast that routine checks could not detect. This did not contribute but increased risk across the fleet. Runway edge lighting measured up to 131% of nominal seven days later. Its effect on detecting the animals could not be determined, because the crew's accounts differed and no measurement from the night exists.

What has been done as a result

The Head of Flying Operations suspended the evening Tarlton Springs service. On 26/06/2026 temporary revision TR-2026-04 listed Tarlton Springs in Appendix 8A, requiring a runway

inspection before every departure there after last light. On 25/06/2026 the aerodrome operator installed a temporary electric wire. It funded the fence repair from its 2026 contingency and programmed replacement of the 140 m section for 09/2026. The investigation recommended that the operator list aerodromes on wildlife exposure rather than a reported strike. It also recommended a route risk assessment trigger for any aerodrome first operated into after last light. No safety action has been recorded for the other safety issues.

Safety message

A wildlife control triggered by a reported strike cannot protect a crew before the first strike. Operators should base such controls on known exposure, including aerodrome operators' sighting logs, and test schedule changes against existing route restrictions. Where a runway inspection is available on request, crews should consider one before departing after last light.

The occurrence

On 18/06/2026 the flight crew of Saltbush Air SB512 were to operate the last scheduled sector of the day from Tarlton Springs Aerodrome. The aircraft was a 50-seat twin-turboprop registered VH-SBQ. The sector was the fifth of the crew's duty. It had been added that morning when the rostered crew reported unfit. The first officer was the pilot flying. The first officer had flown into Tarlton Springs twice before, on 09/01/2026 and 17/02/2026, both times in daylight.

The aerodrome operator's daily movement area inspection was carried out at 21:35 UTC on 17/06/2026, before the first scheduled arrival, and recorded the movement area as clear. No further runway inspection was carried out or requested that day.

Departure and take-off roll

The doors closed at 07:16 UTC, 40 minutes behind schedule, with 44 passengers, 2 flight crew and 1 cabin crew on board. The aerodrome was in darkness. The tower was open and was rostered to close at 18:00 local. The crew calculated a V1 of 112 kt for a take-off weight of 20,800 kg, flap 10, a dry runway 09 and a 5 kt headwind component. The first officer stated that the briefing covered the runway length, the performance figures and the headwind, and that neither pilot raised animals.

At 07:29:10 the tower cleared SB512 for take-off on runway 09, passing the wind as 090 at 6 kt. Neither the tower nor the crew raised a runway inspection. No transmission about wildlife was heard on the tower channel in the preceding 90 minutes.

Take-off power was set at 07:30:02, with the engines at 98.4% and 98.5% torque. The pilot in command called "Power set, both engines good." The 80 kt cross-check followed at 07:30:15, and the pilot in command called V1 at 07:30:29 at 112 kt. No caution, warning or engine parameter exceedance was recorded during the roll, and the two engines remained within 0.1% torque of each other until the power levers were retarded. The recorded headwind component was 4 to 6 kt throughout, with no tailwind component at any point.

The rejected take-off

At 07:30:31, at 116 kt, the first officer called "roos, roos, on the left". Two kangaroos were moving onto the runway from the northern side, past the 1,100 m point. The first officer estimated that they were about 200 m ahead when first seen. She said the range could have been less. She also said the animals were clearly visible once they were on the sealed surface, but that nothing could be seen beyond the lit area before then. After the aircraft stopped, she told the pilot in command on the flight deck: "They came out of the dark on the left of the strip, crossed the centreline." The pilot in command replied: "I never had them. I was heads-in on the engine page after V one."

At 07:30:32 the pilot in command called "Stopping. I have control." The first officer handed over control. Both power levers reached flight idle within 0.6 seconds. The reject began 3 seconds after V1 and at 118 kt, six knots above V1. At that point the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining. The pilot in command later stated that he judged continuing would have put the aircraft airborne at night with damage the crew could not assess. He also stated that he knew the decision was outside the procedure.

About one second later, at 07:30:33, the aircraft struck both animals. The right main landing gear and the No. 2 propeller took the impact. The flight data recorded a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. The No. 2 propeller RPM began to decay from that moment. The carcasses were later recovered 1,188 m and 1,221 m from the runway 09 threshold. Animal tissue was found in two No. 2 propeller blade roots.

Maximum braking of 2,940 psi was established at 07:30:34, 2 seconds after the decision. The operator's fleet mean for rejected take-offs is 3.2 seconds. Anti-skid was active on all four wheels. At 07:30:36 the pilot in command selected reverse. Only No. 1 reverse engaged because the No. 2 reverse interlock did not make. The pilot in command then called "Max brakes, max brakes." The mean deceleration between 07:30:34 and 07:30:44 was 2.6 m/s². That was approximately 15% below the value used in the certified accelerate-stop data, which assumes symmetric reverse. At 07:30:47 the pilot in command said "We're not stopping on it." Two seconds later the first officer called "End's coming up."

The aircraft left the sealed surface at 07:30:51 at 21 kt indicated airspeed. Tyre marks ran continuously from 1,160 m to the runway end. In the graded gravel the deceleration fell to 1.9 m/s². At 07:30:57 the aircraft came to rest 62 m into the 90 m runway end safety area. It was upright and on the extended centreline, with the nose wheel in the gravel. It stopped 28 m short of the end of the prepared safety area, and the terrain beyond that point is ungraded. The stopping distance achieved from 118 kt was 757 m, compared with the 695 m of sealed surface that had remained at the reject.

After the aircraft stopped

At 07:31:04 the pilot in command set the parking brake. He noted no fire indications and nothing on the warning panel, and he stated "I'm not evacuating." The first officer agreed. Both fuel levers were moved to cut-off at 07:31:26. At 07:31:34 the pilot in command told the cabin to remain seated. At 07:31:54, in reply to the tower, he reported the overrun and the persons on board. He requested activation of the aerodrome emergency plan and a vehicle. The tower activated the plan at 07:32:10 and dispatched a vehicle.

At 07:33:02 the pilot in command announced a disembarkation by the forward door. The forward airstair was heard operating at 07:34:40. No evacuation was commanded and no slides were deployed. The main cabin lights remained on throughout. Aerodrome staff attended with a vehicle, and there was no fire. The occurrence notification states that passengers were disembarked 4 minutes 8 seconds after the aircraft stopped. The timeline records the disembarkation as lasting 4 minutes 8 seconds, from 07:34:40 to 07:38:48. [NOT ESTABLISHED: whether the 4 minutes 8 seconds runs from the aircraft stopping or from the start of the disembarkation.]

While walking back through the cabin, the first officer saw unlit sections of the aisle floor proximity strip towards the rear. The emergency lighting was armed at the time and the ceiling lights were on. She reported it to an engineer the next morning. A functional check was carried out on 24/06/2026. [NOT ESTABLISHED: result of the functional check of the floor proximity lighting.]

Injuries, damage and immediate actions

There were no serious injuries. One cabin crew member reported bruising sustained during the disembarkation. Both kangaroos were killed.

The aircraft was substantially damaged. Three No. 2 propeller blades were damaged beyond repairable limits, and the No. 2 engine required a shock-load inspection. The right main gear doors were distorted. Both right main tyres deflated through the fuse plugs after the brakes absorbed 84% of the certified maximum brake energy. The lower right fuselage was abraded and punctured in two places by propeller debris.

The aerodrome operator closed the aerodrome for 4 hours 20 minutes for an inspection and to remove the animal remains and propeller debris. The aircraft was quarantined at Tarlton Springs. The operator's engineering representative removed the FDR and CVR at 12:40 UTC. Both flight crew

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were stood down and tested for drugs and alcohol, with negative results. The Head of Flying Operations suspended the evening Tarlton Springs service.

Context

Flight crew

The pilot in command had 8,410 hours total, 2,140 on type, and had been in command for 3 years 1 month. His last cyclic simulator session (11/03/2026) and line check (02/12/2025) were satisfactory, and he had no prior occurrence history. His duty and rest in the preceding 7 days were within the operator's limits, with no variations applied.

He had seen kangaroos on the strip at Tarlton Springs in the evening on several earlier visits, but never in daylight. He had mentioned this to the aerodrome ground staff. He had never reported a sighting to the operator and was not aware of any other pilot doing so.

The first officer had 1,655 hours total, 405 on type, and 14 months with the operator. Her last cyclic simulator session, on 28/04/2026, was satisfactory. Before the flight she reviewed the flight plan, the charts, the ERSA entry and the operator's aerodrome brief page for Tarlton Springs, and stated that none mentioned wildlife. She had no prior knowledge of kangaroo activity at the aerodrome.

Both crew signed on at 06:30 local. The occurrence was at 17:30 local, 11 hours 00 minutes into a planned duty of 12 hours 05 minutes ending at 18:35 local. The maximum for five sectors from that sign-on was 12 hours 30 minutes. Both crew described themselves as tired at that point but not struggling.

Aircraft

VH-SBQ was a 50-seat twin-turboprop. At 22,900 kg, the type was the heaviest aeroplane scheduled into Tarlton Springs. [NOT ESTABLISHED: aircraft manufacturer, model, year of manufacture and maintenance history before the occurrence.]

After the occurrence, a borescope inspection found no pre-existing damage to either engine and no evidence of a power anomaly before the impact. The anti-skid system tested serviceable, and the right main gear structure was within limits.

The cabin emergency lighting was functionally checked on 24/06/2026 after the first officer reported unlit sections of the aisle floor strip. With the system armed and on battery power, two of the six floor proximity escape path segments, on the left side aft of row 16, did not illuminate. The fault was a failed ballast common to both segments, and the same ballast is fitted across the type.

The type's pre-flight emergency lighting test neither illuminates nor monitors the floor proximity segments, so the defect could not be detected without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month programme interval. Disembarkation was by the forward airstair with the main cabin lights on, and the defect had no effect on it. [NOT ESTABLISHED: serviceability of the floor proximity lighting on other aircraft of the type.]

Meteorology and light

The weather was fine and the runway dry. The observation nearest the departure was:

- "METAR YTLS 180700Z 09006KT 9999 FEW040 08/M01 Q1024 RMK RF00.0/000.0"
[Weather observations]

The 06:30 and 08:00 UTC observations gave 090 at 7 kt and 090 at 5 kt, a headwind component of 5 to 7 kt on runway 09. The crew's forecast gave an easterly of 6 to 8 kt with CAVOK from 10:00 UTC.

On 18/06/2026 sunset at Tarlton Springs was 16:58 local and civil twilight ended at 17:25 local. Moonrise was at 22:41 local, and the moon was a waning crescent 14% illuminated. The take-off roll began at 17:30 local, 32 minutes after sunset. There was no natural ambient light on the strip and no town lighting in sight. On a visit at the same time of day on 23/06/2026, the investigator could not see the unlit graded strip either side of the sealed surface from the runway centreline.

Recorded data

The operator's Flight Data Section read out the FDR and QAR and reported on 24/06/2026. The CVR was transcribed on 25/06/2026 and verified by a second investigator. The tower audio was also transcribed.

The recorded data covered the whole take-off roll, rejected take-off and stop. The power levers reached flight idle 0.6 seconds after the decision, and maximum braking was established 2 seconds after it. This was at the fast end of the operator's fleet distribution, which has a mean of 3.2 seconds from decision to maximum braking. Neither the CVR nor the tower audio contained any request for or offer of a runway inspection, or any mention of wildlife, before departure.

Aerodrome and runway

Tarlton Springs is a certified aerodrome that handled 61,400 passenger movements in 2025. Runway 09/27 is sealed, with an accelerate-stop distance available on runway 09 of 1,847 m. Beyond the runway 09 end is a 90 m runway end safety area of graded gravel, then ungraded terrain.

The daily movement area inspection is carried out before the first scheduled arrival. Aerodrome Manual 6.3 also provides for inspections on request of the reporting officer or a pilot in command, and staff disperse any animals found. The reporting officer confirmed that an inspection would have been carried out if requested on the evening of the occurrence.

A photometric check of the runway edge lighting on 25/06/2026 measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal stage 3 intensity. The others were within tolerance. The maintenance programme checks lamp serviceability but not intensity. No measurement of the lighting on the night of the occurrence exists.

Wildlife management

Aerodrome Standard AS-139 section 10.4 sets the wildlife requirements for certified aerodromes. Subsection 10.4.1 requires the operator to "take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome." Subsection 10.4.2 requires a documented wildlife hazard management plan only "where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year." Under 10.4.3, measures at other aerodromes "need not be documented", but the operator must be able to demonstrate them.

Tarlton Springs was below both limits and was not in breach of the standard. Apart from the Aerodrome Manual 6.3 inspection, it had no documented wildlife measures, including no hazard management plan, hazard assessment or dispersal programme.

Perimeter fence

The aerodrome is operated by Tarlton Shire Council. In 04/2023 the council's aerodrome works register recorded item WR-221: displaced panels and strained wire on the northern boundary fence between the 700 m and 1,300 m markers, with animals seen passing through. Replacement was

estimated at \$41,000 and was deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in 02/2026, recorded further deterioration of the same section.

On 23/06/2026 a continuous 140 m section of the northern fence had its bottom two wires separated from the posts. A clear animal pad ran through the fence at the 1,050 m mark (Figure 1), on the same side as the carcasses and 138 m before the first carcass at 1,188 m.



Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it. Site inspection photograph SV-2026-0417-07, northern boundary at the 1,050 m runway marker, taken 23/06/2026

Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit, showing the fallen section and the animal pad through it (attachments/Perimeter_fence_repair_deferred_for_three_budget_cycles_1.jpg).

Operator procedures and performance data

Route risk assessment and the evening service

Route Risk Assessment RRA-041 issue 3, dated 11/2022, is current for Tarlton Springs. Its hazards include runway length and accelerate-stop margin, the single runway with no lighting redundancy, no alternate within 200 nm at night, and limited engineering support. It carries the restriction "daylight operations only pending review of night lighting and diversion fuel policy."

Change Management Record CMR-2025-118 introduced an evening Sydney to Tarlton Springs rotation, six days a week, from 14/07/2025. It was the operator's first scheduled operation into Tarlton Springs after last light. The record assessed it as "schedule change only", recorded "No new

hazard identified", and against "Route risk assessment review required" recorded "No." The Head of Flying Operations approved it on 22/05/2025.

CMR-2025-118 does not refer to the RRA-041 daylight-only restriction. RRA-041 was not reopened and no issue 4 exists. The Safety Department received only the approved record. The approver has since left the operator.

Wildlife procedures

Operations Manual Part A 8.14.2 states that, at an aerodrome listed in Appendix 8A, the pilot in command "shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed." At other aerodromes, 8.14.3 states that "no specific action is required."

Appendix 8A lists an aerodrome "where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." Part A has no other listing criterion and no provision for crews to report a wildlife sighting. On 18/06/2026 Appendix 8A listed nine aerodromes. Tarlton Springs had never been listed. The list was last reviewed on 03/03/2025, about four months before the evening service began.

Rejected take-off procedure and training

Operations Manual Part B 4.6.3 limits a reject between 80 kt and V1 to "fire or fire warning; engine failure; a configuration warning; loss of directional control; or the pilot in command's judgement that the aircraft is unsafe or unable to fly." Part B 4.6.4 states: "At or above V1 the take-off shall be continued." Section 4.6 has no guidance on a runway obstruction at any speed.

Recurrent training syllabus TRG-SYL-07 module 4 has four scripted simulator scenarios, all aircraft system events: engine failure at 90 kt, engine fire at 100 kt, configuration warning at 85 kt, and engine failure at V1 plus 5 kt with the take-off continued. In the 2025 to 2026 cycle, 214 crew completed the module. Training records for the five cycles on file show no crew presented with an external obstruction scenario. Both pilots independently described their rejected take-off training in the same terms.

Performance

The scheduled accelerate-stop distance required was 1,780 m against 1,847 m available on runway 09, a margin of 67 m. The figure assumes the stop begins at or before V1 with reverse thrust available on both engines.

Wildlife history at the aerodrome

The aerodrome keeps a paper wildlife sighting log in its office. Between 01/2025 and 06/2026 it had 46 entries. Of these, 41 recorded kangaroos on or next to the movement area, 38 of them between one hour before and two hours after last light. Eleven recorded animals on the sealed surface of runway 09/27. The log was not reported to any operator, and the reporting officer confirmed that no airline had ever asked to see it. No wildlife strike involving the operator's aircraft had been reported at Tarlton Springs before the occurrence.

Safety analysis

External Influences

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 section 10.4 sets two levels of obligation for wildlife hazard management. Every certified aerodrome operator "must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome" (10.4.1). A documented wildlife hazard management plan, however, is required only where the aerodrome "is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year" (10.4.2). Below both thresholds, the measures "need not be documented" (10.4.3), although the aerodrome operator must still be able to demonstrate what measures are in place.

Tarlton Springs met neither limb. Its heaviest scheduled aeroplane was the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. It held no wildlife hazard management plan, no hazard assessment and no dispersal programme. Apart from the single inspection before the first arrival under Aerodrome Manual 6.3, none of its wildlife measures were documented. This was confirmed against the certified aerodrome register and the aerodrome operator's own compliance file. The aerodrome was not in breach of the standard. It is virtually certain that the threshold existed and applied to Tarlton Springs in this way.

The factor is therefore treated as an external influence rather than a compliance matter. The aerodrome operator did what the standard required of an aerodrome of its size. The threshold itself, though, left the aerodrome with a wildlife programme that was undocumented and unaudited. In practice, it was also unfunded.

Even so, the aerodrome held information about the hazard. Its wildlife sighting log for 01/2025 to 06/2026 contained 46 entries, of which 41 related to kangaroos on or adjacent to the movement area. Of those 41, 38 fell between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked for it.

The condition of the northern perimeter fence was also known, as the works register shows. Item WR-221, raised in 04/2023, recorded the fence as failed between the 700 m and 1,300 m runway markers, with animals observed passing through. Item WR-263, raised in 02/2026, recorded the same section as deteriorating further. A site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts and an animal pad through it at the 1,050 m mark.

The case for influence rests on what a documented plan would have provided. A plan is the mechanism that would have connected the aerodrome's sighting log to the operators using the aerodrome. It would also have placed the fence repair on a funded programme instead of an unfunded works register. A documented plan brings an audit trail and a person responsible for it, and every other control that failed in this occurrence did so for want of those two things.

There is also evidence against this view. Section 10.4.1 still required the aerodrome operator to take reasonable measures, and a documented plan is not the only way to repair a fence or to pass a sighting log to an airline. The council's deferrals of the fence repair were budget decisions, and no evidence showed that a documented plan would have been funded any better. A plan is a document, whereas the fence repair needed money.

On the balance of this evidence, the investigation found that the threshold likely contributed to the occurrence. This conclusion is less firm than the other contributing factors in this analysis, because it depends on what a documented plan would have changed rather than on direct evidence that it would have changed the outcome. The argument that funding, not documentation, determined the fate of the fence was not disproved.

The threshold influenced the occurrence through two other factors. The deferral of the perimeter fence repair across three budget cycles is discussed under Organisational Influences. The requirement for a runway inspection only at listed aerodromes, where the listing criterion is a strike, is discussed under Risk Controls.

Organisational Influences

Perimeter fence repair deferred for three budget cycles

The aerodrome operator's works register shows that the northern perimeter fence was known to have failed more than three years before the occurrence. Item WR-221, raised in 04/2023, recorded four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers, and noted that animals had been seen passing through. Replacement was recommended at an estimated cost of \$41,000. The item was deferred as unfunded at the 2023, 2024 and 2025 budget rounds, and each deferral carried it forward for review at the next cycle. In 02/2026, item WR-263 recorded that the same section was deteriorating further.

The site inspection on 23/06/2026 found the fence in that condition. A continuous 140 m section had the bottom two wires separated from the posts, and a clear animal pad ran through the fence at the 1,050 m mark (Figure 1). The runway survey recovered the two carcasses at 1,188 m and 1,221 m from the runway 09 threshold, on the same side of the runway as the breach. The first officer saw the animals moving onto the runway from the northern side, past the 1,100 m point, and crossing towards the south.

It was virtually certain that the fence had been left unrepaired across three budget cycles. The aerodrome operator's own register documents the deferrals, and the site inspection confirmed the condition of the fence. A fence in that condition does not exclude kangaroos. The animal pad through the breach lay directly upstream of the strike position, so the breach was the physical route by which the animals reached the movement area. On that basis it was very likely that the deferred repair contributed to the two kangaroos entering the runway strip and moving onto the sealed surface.

The aerodrome's wildlife sighting log for 01/2025 to 06/2026 shows how much the aerodrome operator knew about this exposure. Of its 46 entries, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and 11 recorded animals on the sealed surface of runway 09/27. The log was a paper register held in the aerodrome office, and it was not reported to any operator. The aerodrome reporting officer confirmed that no airline had ever asked to see it. The hazard was therefore recorded at the aerodrome in two places, the works register and the sighting log. Neither record led to the fence being repaired, and the information did not reach the operator that was flying there after last light.

The deferrals took place under a regulatory framework that did not require a documented wildlife hazard management plan at an aerodrome of this size. That condition is discussed under External Influences. The timing of the departure in relation to kangaroo movement is discussed under Local Conditions.

Evening service approved as a schedule change, route risk assessment not reopened

The operator's current route risk assessment for Tarlton Springs is RRA-041 issue 3, dated 11/2022. It assessed hazards including "runway length and accelerate-stop margin" and "single runway with no lighting redundancy", and it imposed the operating restriction "daylight operations only pending review of night lighting and diversion fuel policy". No issue 4 exists.

Change Management Record CMR-2025-118 introduced "one evening rotation, Sydney to Tarlton Springs to Sydney, six days per week, from 14/07/2025". This was the first scheduled company operation into Tarlton Springs after last light. Even so, the record assessed the change as "schedule change only" and stated "No new hazard identified." Against route risk assessment review it recorded "No." The Head of Flying Operations approved it on 22/05/2025. The record does not refer to the daylight-only restriction anywhere, and RRA-041 was not reopened. The Safety Department's records show no involvement in the change beyond receipt of the approved record.

It was virtually certain that the evening service was approved as a schedule change without the route risk assessment being reopened. The documents are current controlled copies, and nothing contradicts them. Why the change was assessed in this way could not be established, because the approver has left the company and could not be interviewed. The report therefore draws no conclusion about what the approver considered or intended. What the documents do show is that the change process asked whether the operation was new in kind. It did not ask whether the existing route risk assessment still held for an operation that the assessment itself restricted.

A review of RRA-041 against a night operation would have reached two matters: the aerodrome's wildlife exposure after last light, and whether Tarlton Springs should be listed in Appendix 8A of the Operations Manual. Listing would have required a runway inspection before the first departure after last light. Two considerations weigh against certainty. The restriction names only night lighting and diversion fuel policy, so a review might have dealt with those matters alone. In addition, the approver could not be asked what information was in front of him. On balance, it was likely that the approval without review contributed to the absence of any requirement for a runway inspection at Tarlton Springs.

The inspection requirement in the Operations Manual depended on a reported strike to list an aerodrome, and Tarlton Springs had never been listed. That gap, and the resulting absence of a runway inspection before the departure, are discussed under Risk Controls.

Risk Controls

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

The operator's Operations Manual Part A contained a control directed at the hazard in this occurrence. Section 8.14.2 requires that at an aerodrome listed in Appendix 8A, "the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed." At every other aerodrome, section 8.14.3 states that "no specific action is required." Because the manual text and the appendix listing are documentary, it is virtually certain that the control existed in this form.

The weakness lay in how an aerodrome came to be listed. Appendix 8A admits an aerodrome only "where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." No provision in Part A admits an aerodrome on any other evidence, and none asks a crew to report a sighting. Section 8.14.4 ties a review of an aerodrome's status to a strike. As a result, the inspection requirement could apply at an aerodrome only after the outcome it existed to prevent had already happened there. Nine aerodromes were listed. Tarlton Springs

had never been listed, and the list was last reviewed on 03/03/2025, four months before the evening service began.

The exposure at Tarlton Springs was documented locally. The aerodrome's paper wildlife sighting log for 01/2025 to 06/2026 contained 46 entries, and 41 of these recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was kept in the aerodrome office and was not reported to any operator. None of these entries was a strike, so none of them could place the aerodrome on the list.

It is very likely that the listing criterion contributed to the absence of a runway inspection before this departure. Had section 8.14.2 applied, an inspection would have been requested and its result passed before the roll, and aerodrome staff disperse animals when they find them. The assessment is no stronger than very likely because an inspection completed some minutes before departure does not guarantee a clear runway at the start of the roll. The absence of an inspection request on the night is discussed under Individual Actions. The external and organisational conditions that preceded this criterion are discussed under their own headings.

Rejected take-off training covers aircraft system failures below V1 only

Part B 4.6 sets the standard against which the reject is measured. Section 4.6.4 states: "At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached." The section gives no guidance on an obstruction seen on the runway at any speed, and it does not address a case in which continuing is itself hazardous. Module 4 of TRG-SYL-07 matches this. Its four scripted scenarios were an engine failure at 90 kt, an engine fire at 100 kt, a configuration warning at 85 kt and an engine failure at V1 plus 5 kt, continued. Each is an aircraft system event. In the five training cycles held by the company, no crew had been presented with an external obstruction scenario, and 214 crew completed the module in the 2025 to 2026 cycle. Both pilots stated independently that every stop or go scenario they had been given was an engine event below V1. It is extremely likely that this gap existed. The assessment falls short of virtually certain only because scenario details from before those five cycles were not available.

The question of influence was argued in both directions. The first officer's sighting call was made at 07:30:31 at 116 kt, and the reject began at 07:30:32 at 118 kt. The first officer described having the animals in view for "two seconds. Maybe less," and the captain said, "I never had them." On one reading, the captain met the question of an obstruction above V1 for the first time in that interval, with no prepared frame to draw on. On the other, the captain knew the rule and departed from it deliberately because he judged continuing to be the worse option. On that view, training on obstructions would not have changed what he knew, and the gap would be a fleet-wide matter rather than a cause of this reject.

Both readings are retained. A decision made without rehearsal in about a second is not the same as one made against a briefed and practised frame. On that basis it is likely that the training gap contributed to the take-off being rejected 6 kt above V1. This explains the conditions under which the decision was made, but it does not set aside section 4.6.4. The reject itself is discussed under Individual Actions.

Cabin floor proximity emergency lighting partially unserviceable

The first officer noticed unlit sections of the aisle floor strip while walking back through the cabin during the disembarkation. A functional check on 24/06/2026, with the system armed and the aircraft on battery power, found that two of the six floor proximity escape path segments on the left side aft of row 16 did not illuminate. The fault was traced to a failed ballast common to both segments. It is virtually certain that the defect was present.

It is unlikely that the defect influenced the occurrence or its outcome. No evacuation was commanded, the main cabin lights were on throughout, and passengers left without urgency by the forward airstair.

The defect is still a safety issue, and it is very likely that it increased risk. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs, but it neither illuminates nor monitors the floor segments. Neither the crew nor a line engineer could therefore have detected the fault without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month programme interval. In a night evacuation with fire or smoke, or over the wing, dark segments in the aft cabin could impede the exit. Because the same ballast is fitted across the type, the condition of the floor proximity lighting on other aircraft in the fleet was not known.

Local Conditions

Departure in the hour after last light, the peak of kangaroo movement

The take-off roll began at 17:30 local time, 32 minutes after sunset and 5 minutes after the end of civil twilight. The moon would not rise until 22:41 and was 14% illuminated. There was no natural ambient light on the strip and no town lighting within sight. At a site visit at the same time of day on 23/06/2026, the investigator found that the unlit graded strip either side of the sealed surface could not be seen from the centreline.

The aerodrome's wildlife sighting log placed this departure inside its period of greatest kangaroo activity. From 01/2025 to 06/2026 the log held 41 entries recording kangaroos on or adjacent to the movement area. Of these, 38 fell between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The astronomical data and the log are documentary, but the log is a paper register whose completeness could not be verified. On that basis it was extremely likely that the departure took place in the peak period of kangaroo movement.

It was likely that this condition influenced the occurrence. The first officer did not see the animals until they were on the lit sealed surface. On the CVR she said they "came out of the dark on the left of the strip". That account matches what the site visit showed, since an animal on the graded strip at that time of day could not be seen from the aircraft. The timing also coincided with the movement peak in the aerodrome's own log.

Taken alone, a departure after last light is unremarkable. Its significance lay in two things occurring together: the hours of greatest wildlife movement, and a light condition in which nothing off the sealed surface was visible. That combination is what gave the absence of a runway inspection its weight. The inspection control is discussed under Risk Controls.

Runway edge lighting intensity reduced the crew's ability to detect the animals

The investigation considered whether the runway edge lighting reduced the crew's ability to see the animals. One fact was settled. A photometric check on 25/06/2026 measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal intensity for stage 3. These lights were in the section the animals crossed, and the remaining lights were within tolerance. The aerodrome's maintenance programme checks lamp serviceability but does not measure intensity. The open question was whether that deviation reduced detection on the night, and the evidence left this about as likely as not.

In support, the captain recalled the lighting as brighter than he would want. He described a corridor of light with everything outside it black, and said that anything off the seal could not be seen coming.

The first officer's account pointed the other way. She was the crew member who saw the animals, and she described the lighting as normal. She said she saw them clearly once they were on the sealed surface, at around 200 m. She also said that with only two daylight landings at Tarlton Springs to compare against, she was not well placed to judge whether the lighting was unusual. The captain's own failure to see the animals has an explanation that does not depend on the lighting. On the CVR he said, "I never had them. I was heads-in on the engine page after V one." The procedure placed his attention there.

The matter could not be resolved, for three reasons. The photometric check was made seven days after the occurrence, and nothing showed what the lights were doing on the night. The two crew accounts disagreed. No available evidence linked an edge light at 131% of nominal to a reduced ability to detect an object on the runway. Resolving the matter would need both a measurement of the lighting as it was on the night and evidence of how that intensity affects detection of an object on the sealed surface. The matter remains unresolved, and no finding is made about it.

Take-off commenced with a tailwind above limits

A tailwind was ruled out as very unlikely. The aerodrome observations of 090 at 5 to 7 kt and the FDR, which recorded a headwind component of 4 to 6 kt throughout the roll and no tailwind at any point, both agreed with the 5 kt headwind used in the crew's performance calculation.

Fifth sector, 11 hours into duty, 40 minutes behind schedule

It was virtually certain that the occurrence sector was the fifth of a duty that began at 06:30 local. The fifth sector had been added that morning when the rostered crew reported unfit. The take-off was 11 hours into a duty planned to last 12 hours 05 minutes, against a limit of 12 hours 30 minutes, and the aircraft departed 40 minutes behind schedule. Both crew described themselves as tired but not struggling.

Degraded decision making would be expected this late in a long duty. Even so, it was unlikely that the duty affected the crew's actions in this occurrence. The FDR showed 0.6 seconds from the decision to power levers at flight idle, and 2 seconds to maximum braking. That response was at the fast end of the operator's fleet distribution, where the fleet mean was 3.2 seconds, and no step was omitted on the roll. The duty might plausibly have affected one decision, whether to request a runway inspection. However, no document prompted either crew member to consider that step, so it was not a decision either of them was asked to make.

The duty pattern was nonetheless likely to have increased risk. The evening Tarlton Springs rotation routinely places the last sector in the final hour of a 12-hour duty. The aerodrome has no company engineering and no alternate within 200 nm at night. With no crewing resilience, a single unfit crew member turns a four-sector day into a five-sector day on the morning of operation, and that exposure recurs whenever a crew member reports unfit.

Individual Actions

No runway inspection requested before the first departure after last light

No runway inspection was requested before SB512 departed, and none was offered. The CVR and the tower's recorded audio contain no request for an inspection, no offer of one and no mention of wildlife at any point before departure. The aerodrome's daily inspection log for 18/06/2026 shows a single inspection, at 07:35 local before the first arrival, which found the movement area clear. No later inspection was carried out or requested. It is virtually certain that no inspection took place before the first departure after last light.

An inspection was available on request. Aerodrome Manual 6.3 provides for inspections outside the scheduled time when the reporting officer or a pilot in command asks for one, and the reporting officer confirmed that one would have been carried out if asked. Both crew members stated that they did not consider requesting one. The captain had expected Tarlton Springs to be on the Appendix 8A list and was surprised to find it was not. At an aerodrome not on that list, section 8.14.3 states that "no specific action is required." Nothing in the operator's procedures prompted either pilot to make the request, so the inspection was not omitted against any instruction. The reason Tarlton Springs could not be listed is discussed under Risk Controls.

An inspection would have covered the section of runway 09 from 1,050 m to 1,250 m, and both carcasses were later recovered within that section, at 1,188 m and 1,221 m from the threshold. Aerodrome staff disperse animals when they find them. The aerodrome's own sighting log showed that kangaroos were regularly present at this time of day: of 41 kangaroo entries, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface. Against this, an inspection completed some minutes before the take-off roll could not guarantee that the runway would stay clear, because animals could return to it afterwards. On balance, the absence of an inspection likely contributed to the two kangaroos being on the runway when the aircraft reached them.

Take-off rejected 6 kt above V1

The captain called V1 at 112 kt at 07:30:29. Two seconds later, at 07:30:31, the first officer called kangaroos on the left. At 07:30:32 the captain announced "Stopping. I have control." The FDR shows both power levers moving to flight idle within 0.6 seconds, at 118 kt, three seconds and 6 kt after V1. Engine parameters on both engines were normal up to that point, so the reject was a response to the animals alone. It is virtually certain that the take-off was rejected above V1.

Part B 4.6.4 states: "At or above V1 the take-off shall be continued." Section 4.6 gives no guidance on an obstruction seen on the runway at any speed, and none for a case in which continuing is itself hazardous. The captain gave his account in interview and in his air safety report. He judged that rotating into two large animals would leave the aircraft airborne at night with damage he could not assess, over country with no usable option for a return. He chose the risk whose end he could see, and he acknowledged that the decision was not in accordance with procedure. The decision was made and announced in a single exchange, about one second after the sighting call. The CVR also shows that the captain had not seen the animals himself. After the aircraft stopped, he said, "I never had them. I was heads-in on the engine page after V one."

The scheduled accelerate-stop distance of 1,780 m assumes that the reject begins at or before V1, and the margin between the distance required and the distance available was 67 m. At the moment of the reject the aircraft was 1,152 m along runway 09, with 695 m of sealed surface remaining. The stopping distance achieved from 118 kt was 757 m. Beginning the stop six knots and three seconds late therefore used up the 67 m margin and more, even before the loss of No. 2 reverse is taken into account. It is extremely likely that the late reject contributed to the overrun.

The decision did not arise in isolation. Two factors lie upstream of it: the animals being on the runway at all, and the operator's rejected take-off training covering only aircraft system failures below V1. Those factors are discussed under Local Conditions and Risk Controls. Whether the runway edge lighting affected how early the animals could be seen is discussed as an unresolved matter under Local Conditions.

Maximum braking and No. 1 reverse applied without delay

Once the stop was committed, the crew carried it out quickly. The power levers reached flight idle 0.6 seconds after the decision, and maximum braking was established at 07:30:34, 2 seconds after

it. The operator's recorded fleet mean from decision to maximum braking is 3.2 seconds, which places this response at the fast end of the fleet distribution. Anti-skid was active on all four wheels throughout. Brake energy reached 84% of the certified maximum, and there was no brake fire. The captain selected No. 1 reverse at 07:30:36. No. 2 reverse was not available because the No. 2 propeller had been damaged in the strike. Meanwhile the first officer called the speeds and warned that the runway end was approaching. It is virtually certain that this response occurred as described.

Mean deceleration on the sealed surface was 2.6 m/s², approximately 15% below the value in the certified accelerate-stop data, which assumes symmetric reverse. The aircraft left the seal at 21 kt and decelerated at 1.9 m/s² in the gravel. It stopped at 07:30:57, 62 m into the 90 m RESA, upright and on the extended centreline. Had the stop begun at the fleet mean response time, the aircraft would have travelled an estimated further 30 to 40 m, close to the far edge of the RESA. The prompt braking and reverse very likely reduced the severity of the overrun. Its outcome was a low-speed stop on a prepared surface rather than a departure from the RESA at speed. The passengers were then disembarked by the forward airstair without an evacuation.

Occurrence Events

Two kangaroos entered the runway strip and moved onto the sealed surface

It is virtually certain that two kangaroos entered the runway strip and reached the sealed surface of runway 09. Three independent sources each establish this on their own. Two adult kangaroo carcasses were recovered from the northern half of the runway, at 1,188 m and 1,221 m from the threshold. The CVR records the first officer's sighting call at 07:30:31: "Roos - roos, roos, on the left, they're on the -". The FDR and the CVR both record an impact at 07:30:33.

The first officer saw the animals crossing from the northern side towards the south and estimated their range at first sight at around 200 m. Immediately after the aircraft stopped, she described them as having come "out of the dark on the left of the strip". She said they were visible for about two seconds, "maybe less". The captain did not see them.

The physical evidence places the animals close to a known breach in the perimeter fence. The site inspection found an animal pad through the failed northern fence at the 1,050 m mark, 138 m before the first carcass and on the same side of the runway. The hazard was not new, as the aerodrome's wildlife sighting log for 01/2025 to 06/2026 shows. Of 41 kangaroo entries, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. No runway inspection had been carried out since 07:35 local that morning.

The analysis here concerns only whether the animals were on the runway. The route by which they reached it bears on the deferred fence repair, which was found to have contributed and is considered under Organisational Influences. The timing of the departure relative to last light, and the absence of a runway inspection, are considered under Local Conditions and Individual Actions.

It is extremely likely that the presence of the animals on the sealed surface contributed to the overrun. The rejected take-off and the strike both followed directly from it. The reject decision came one second after the sighting call, and the impact came one second after that.

No. 2 propeller and right main gear struck two kangaroos

It is virtually certain that the aircraft struck both animals, and virtually certain that the strike contributed to the overrun. The recorded data, the physical damage and the recovered tissue all agree. The impact occurred at 07:30:33 at 114 kt, one second after the reject began. The FDR recorded a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. Three No. 2 propeller blades were damaged beyond repairable limits, one blade tip had separated, and

animal tissue was recovered from two blade roots. The right main gear doors were distorted, and propeller debris punctured the lower right fuselage skin.

The strike did more than damage the aircraft. No. 2 propeller RPM decayed from the moment of impact, and the No. 2 reverse interlock did not make. When reverse was selected at 07:30:36, it was available on No. 1 only. The captain's call at that moment was "Reverse - number two's not coming." The strike therefore removed one of the two reverse units about two seconds into a stop that had a scheduled margin of 67 m. That loss of reverse thrust is why the aircraft left the end of the runway rather than stopping on it.

Deceleration achieved below the scheduled accelerate-stop performance

It is virtually certain that the aircraft decelerated at less than the scheduled accelerate-stop performance, and virtually certain that this shortfall contributed to the overrun. The recorded data and the distance reconstruction agree with the tyre marks and the nose wheel rut.

The scheduled accelerate-stop distance required was 1,780 m against an ASDA of 1,847 m, a margin of 67 m. That figure assumes a stop begun at or before V1 with symmetric reverse, and neither assumption held. Distance reconstruction from recorded groundspeed placed the aircraft 1,152 m along runway 09 when the reject began at 118 kt, leaving 695 m of sealed surface. The stopping distance actually achieved from that speed was 757 m. The mean deceleration on the sealed surface was 2.6 m/s², approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition. After the aircraft left the seal at 21 kt, deceleration fell to 1.9 m/s² in the gravel, and the aircraft stopped 62 m into the RESA.

The shortfall is a single mechanism with two inputs. First, the stop began 6 kt above V1, which is considered under Individual Actions. Second, it was carried out with about 15% less deceleration than the figures assume, because the strike removed No. 2 reverse. The 62 m overrun is the combined result of both, and neither input alone carried the aircraft beyond the runway end.

No. 2 engine power loss during the take-off roll

A No. 2 engine power loss before the reject was very unlikely. Both engines held take-off torque within 0.1% of each other, with no caution or warning, until the power levers were retarded at 07:30:32. The No. 2 RPM decay began only at the impact, one second later.

Findings

From the evidence available, the following findings are made with respect to the rejected take-off and runway overrun involving VH-SBQ at Tarlton Springs Aerodrome on 18/06/2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

A finding stated without a likelihood term was established as fact.

Contributing factors

- With virtual certainty, Aerodrome Standard AS-139 section 10.4 required a documented wildlife hazard management plan only above a size threshold that Tarlton Springs did not meet. This likely contributed by leaving the aerodrome's wildlife measures without an owner or an audit trail. That influence is less firmly established than the other contributing factors, because a documented plan would not of itself have funded the fence repair. (Safety Issue)
- It is virtually certain that the repair of the failed northern perimeter fence was deferred for three budget cycles, and very likely that this contributed by leaving an animal route through the fence 138 m before the first carcass position, on the same side of the runway. (Safety Issue)
- It is very likely that the departure took place in the hour after last light, during the peak of kangaroo movement recorded at the aerodrome, and likely that this contributed, because animals on the unlit strip could not be seen until they reached the lit sealed surface.
- The operator's evening service to Tarlton Springs was, with virtual certainty, approved as a schedule change without route risk assessment RRA-041 and its daylight-only restriction being reopened. This likely contributed by leaving the aerodrome's wildlife exposure and its Appendix 8A status unexamined before night operations began, although such a review might have addressed only the lighting and fuel matters the restriction named. (Safety Issue)
- It is virtually certain that Operations Manual Part A 8.14 required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A, and that the only listing criterion was a reported wildlife strike. It is very likely that this contributed, because Tarlton Springs was not listed despite eleven recorded sightings of kangaroos on the sealed surface. (Safety Issue)
- It is virtually certain that no runway inspection was requested before the first departure after last light, an omission that no procedure prompted either pilot to consider. It is likely that this contributed, as an inspection would have covered the section where the animals were struck and aerodrome staff disperse animals they find, although an inspection some minutes before the roll could not guarantee a clear runway.
- Two kangaroos entered the runway strip and moved onto the sealed surface during the take-off roll, which is virtually certain. This very likely contributed by prompting the rejected take-off and placing the animals in the aircraft's path.
- The operator's rejected take-off training very likely covered only aircraft system failures, three of its four scenarios below V1, with no external obstruction scenario. This likely contributed by leaving the pilot in command to decide whether to reject above V1 without any prepared frame for that decision. The pilot in command also stated that he knew the procedure and departed from it deliberately. (Safety Issue)

- The recorded data establish with virtual certainty that the take-off was rejected at 118 kt, 6 kt and 3 seconds after V1. It is very likely that this contributed, because the scheduled accelerate-stop distance assumed a stop beginning at or before V1 and the 67 m margin was consumed.
- With virtual certainty, the No. 2 propeller and right main gear struck two kangaroos one second after the reject began. This contributed with virtual certainty by leaving reverse available on No. 1 only during the stop.
- It is virtually certain that the deceleration achieved was below the scheduled accelerate-stop performance, at 2.6 m/s² or approximately 15% below the certified value. It is also virtually certain that this, together with the late start of the stop, resulted in the aircraft leaving the runway at 21 kt and stopping 62 m into the runway end safety area.

Other factors that increased risk

- Two of the six cabin floor proximity escape path segments on VH-SBQ were, with virtual certainty, dark because of a failed common ballast that the pre-flight emergency lighting test could not detect. It is very likely that this increased risk, as the same ballast is fitted across the type and the 24-month functional test interval cannot reveal such a defect between tests. (Safety Issue)
- With virtual certainty, the occurrence sector was the crew's fifth, 11 hours into duty and 40 minutes behind schedule. It is likely that this increased risk, because the evening Tarlton Springs rotation routinely places the last sector near the end of a 12-hour duty and a single unfit crew member can add a sector on the morning of operation.

Other findings

- It is very unlikely that the take-off commenced with a tailwind above limits, as the aerodrome observations and the recorded data both showed a headwind component throughout the roll.
- A No. 2 engine power loss during the take-off roll was very unlikely, as both engines were within 0.1% torque of each other until the reject, and the No. 2 propeller RPM decay began at the moment of impact.
- With virtual certainty, maximum braking was applied 2 seconds after the decision to reject, against a fleet mean of 3.2 seconds, and No. 1 reverse was applied without delay. This very likely reduced the extent of the overrun, which a stop begun at the fleet mean response would have lengthened by an estimated further 30 to 40 m.

Safety issues and actions

Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 section 10.4 required a documented wildlife hazard management plan only at an aerodrome used by aeroplanes with a maximum take-off weight above 27,000 kg, or serving more than 350,000 passenger movements a year. Tarlton Springs met neither limb, so its wildlife measures did not have to be documented. The aerodrome was not in breach of the standard.

The issue matters because of what the threshold left out. Without a documented plan, the aerodrome's wildlife measures had no owner and no audit trail. The aerodrome's own sighting log recorded 41 kangaroo entries between 01/2025 and 06/2026. Of these, 38 were around last light and eleven were on the sealed surface. That log was kept as a paper register that no operator had ever asked to see. The failed northern perimeter fence stayed on an unfunded works register rather than a funded programme. Through these effects the threshold likely contributed to the occurrence. It led to the deferral of the fence repair and to a runway inspection requirement that depended on a reported strike.

That influence is less firmly established than the other contributing factors. Section 10.4.1 still obliged the aerodrome operator to take reasonable measures. A documented plan is also not the only means of repairing a fence or passing a sighting log to an airline. There is no evidence that a documented plan would have been funded any better.

No safety action has been recorded in response to this safety issue.

Perimeter fence repair deferred for three budget cycles

It is virtually certain that repair of a failed section of the northern perimeter fence at Tarlton Springs was deferred for three budget cycles. Works register item WR-221, raised in 04/2023, recorded four displaced panels and strained wire between the 700 m and 1,300 m runway markers, with animals observed passing through. Replacement was estimated at \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds, and each entry read "defer, unfunded, review next cycle". Item WR-263, raised in 02/2026, recorded the same section deteriorating further.

The site inspection on 23/06/2026 found a continuous 140 m section with the bottom two wires separated from the posts. A clear animal pad ran through the fence at the 1,050 m mark (Figure 1). The carcasses were recovered at 1,188 m and 1,221 m, on the same side of the runway. A fence in that condition does not exclude kangaroos. The breach was very likely the route by which the two animals reached the movement area. The aerodrome operator's own register showed that both the breach and animal movement through it had been known since the first entry in 2023.

Figure 1: The northern perimeter fence at the 1,050 m mark, photographed on the site visit: the fallen section and the animal pad through it

attachments/Perimeter_fence_repair_deferred_for_three_budget_cycles_1.jpg

Proactive safety action

Action taken by the Tarlton Springs aerodrome operator

The aerodrome operator funded works register item WR-221 from the 2026 contingency. It programmed replacement of the 140 m northern boundary fence for 09/2026. A temporary electric wire was installed on 25/06/2026. The action was recorded as in progress.

Evening service approved as a schedule change, route risk assessment not reopened

The operator approved its evening service to Tarlton Springs as a schedule change without reopening the route risk assessment that restricted the aerodrome to daylight operations. Route risk assessment RRA-041 issue 3 carried the restriction "daylight operations only pending review of night lighting and diversion fuel policy". Change Management Record CMR-2025-118 introduced a scheduled evening rotation from 14/07/2025 and assessed it as "schedule change only". The record answered "No" against route risk assessment review and did not mention the restriction. The Head of Flying Operations approved the change on 22/05/2025. RRA-041 was not reopened and no later issue exists.

This issue matters because the change put a company aircraft into Tarlton Springs after last light for the first time. It did so against a restriction still current in the operator's own route risk assessment. The change process asked whether the operation was new. It did not ask whether the existing assessment still held. A review of RRA-041 against a night operation would likely have reached the aerodrome's wildlife exposure and its absence from Appendix 8A. Such a review might, however, have addressed only the lighting and fuel matters the restriction named. Without a step that tests a change against existing restrictions, other operational changes could pass through the same process unexamined.

No safety action has been recorded in response to this safety issue.

Runway inspection required only at listed aerodromes, and the listing criterion is a strike

It is virtually certain that Operations Manual Part A 8.14.2 required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A. An aerodrome could be listed only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. Section 8.14.3 required nothing at other aerodromes. No provision asked a crew to report a sighting, and no other evidence could admit an aerodrome to the list. Nine aerodromes were listed. Tarlton Springs never had been.

The inspection requirement was the operator's own defence against this kind of occurrence, but it could take effect at an aerodrome only after a strike had happened there. The Tarlton Springs wildlife sighting log for 01/2025 to 06/2026 held eleven entries of kangaroos on the sealed surface. None of them could list the aerodrome, because they were not strikes and were not reported to the company. The list was last reviewed on 03/03/2025, four months before the evening service began. It is very likely that this criterion contributed to the occurrence.

On 26/06/2026 the Head of Flying Operations amended Appendix 8A by temporary revision TR-2026-04 to list Tarlton Springs. The Part A 8.14.2 inspection requirement therefore applies to every departure there after last light, pending a permanent change. This action is closed.

Proposed safety action: The investigation recommends that the operator replace the Appendix 8A listing criterion so that an aerodrome is listed on its wildlife exposure rather than on a reported strike. It also recommends that the operator add a route risk assessment trigger for any aerodrome first operated into after last light. A response was due by 30/09/2026, and the recommendation is open.

Rejected take-off training covers aircraft system failures below V1 only

The operator's rejected take-off training and procedures were extremely likely to have prepared crews only for aircraft system failures, and gave no guidance on an obstruction on the runway.

Module 4 of TRG-SYL-07 contained four scripted scenarios, all of them aircraft system events. Three of these occurred below V1, and the fourth was an engine failure at V1 plus 5 kt in which the take-off was continued. No crew had been presented with an external obstruction scenario in the five training cycles held by the operator. In the 2025 to 2026 cycle alone, 214 crew completed the module. Operations Manual Part B section 4.6 matched the syllabus. It required the take-off to be continued at or above V1 and contained no guidance on an obstruction on the runway at any speed.

The issue matters because a crew facing an obstruction near or above V1 must make the stop or go decision without a briefed or practised frame. It is likely that this contributed to the occurrence. The pilot in command decided to reject within about two seconds, and it was the first time he had considered that question. However, he also stated that he knew the procedure and departed from it deliberately because he judged the alternative to be worse. On that view, obstruction training would not have changed what he knew, and the gap bears more on the fleet than on this reject. Both readings are carried here. The issue explains the conditions in which the decision was made and does not excuse the decision.

No safety action has been recorded in response to this issue.

Cabin floor proximity emergency lighting partially unserviceable

It is virtually certain that two of the six cabin floor proximity escape path segments on VH-SBQ were dark because of a failed common ballast. The segments were aft of row 16 on the left side. The first officer noticed sections of the aisle floor strip unlit while walking back through the cabin during the disembarkation. A functional check on 24/06/2026 confirmed that the two segments did not illuminate on battery power with the system armed.

The defect played no part in the overrun or its outcome. No evacuation was commanded and the main cabin lights were on throughout. The disembarkation, by the forward airstair, took 4 minutes 8 seconds without incident.

It is very likely that the defect nonetheless increased risk. The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs, but it neither illuminates nor monitors the floor segments. Neither the crew nor a line engineer could have detected the failure without a dedicated functional test. The last such test was 11 months before the occurrence, within the 24-month programme interval. That interval cannot reveal a defect of this kind between tests. Because the same ballast is fitted across the type, the condition of the floor proximity lighting on the rest of the fleet was not known. In an evacuation involving a fire indication, smoke or a night exit over the wing, two dark segments in the aft cabin could turn an orderly exit into a blocked one. (Safety Issue)

No safety action has been recorded in response to this safety issue.

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